Isomeric States in ^{269}Hs and the Shell Structure near $N = 162$ Probed via ^{273}Ds Decay

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Abstract

An extensive series of experiments was performed at the SHANS-2 separator at IMP CAS. First, the cross section of the ^{238}U(^{40}Ar, 4-5n)^{274,273}Ds reaction was measured in the excitation-energy range from 42 to 50 MeV. Previously, the same reaction was studied in a JINR-IMP CAS collaborative experiment at the DGFRS-2 separator at an energy of 49 MeV. In those prior experiments, two decay chains of ^{273}Ds were observed, exhibiting decays with different energies and lifetimes, which indicated the existence of an isomeric level in addition to the ground state. In the present SHANS-2 experiments, the excitation energies of 42, 45, and 50 MeV were studied in detail, establishing upper cross-section limits of 0.34, 1.09 pb, respectively. However, at 50 MeV, two decay chains of ^{273}Ds were measured, yielding a cross section of $0.14^{+0.18}_{-0.08}$ pb, which is consistent with the $0.18^{+0.44}_{-0.12}$ pb measured at DGFRS-2. Notably, the isomer chains observed at SHANS-2 represent previously unknown decay pathways. Furthermore, by analyzing the α-particle energy branching ratios of ^{269}Hs produced from different sources, the possible existence of an isomeric state in ^{269}Hs is discussed.

Keywords: α Decay, $N = 162$ subshell, Isomeric state, Gas-filled recoil separator

1. Introduction

The macroscopic liquid-drop model precludes the existence of superheavy elements (SHEs, $Z \geq 104$) due to dominant Coulomb repulsion . Nevertheless, remarkable experimental and theoretical advances have enabled the synthesis of nuclei in this extreme mass region,especially with ^{48}Ca-induced reactions [1, 2]. The very existence and survival of these SHEs rely intrinsically on quantum shell effects, which supply the requisite additional binding energy to stabilize them against spontaneous fission. Theoretical models predict that these shell corrections manifest as "islands of stability" across the superheavy landscape. Enhanced stability is anticipated near the deformed doubly magic closures at $Z = 108$ and $N = 162$ [3], extending toward heavier spherical configurations near $N = 184$ and $Z = 114 - 126$, albeit with model-dependent variations[4].Probing odd-A nuclei in this mass region offers a stringent test of these theoretical frameworks, yielding direct access to the single-particle levels that govern the deformed $N = 162$ shell closure. Specifically, the odd-neutron nucleus ^{273}Ds constitutes a single-particle state just above the $N = 162$ gap, while its α-decay daughter, ^{269}Hs, acts as a one-neutron hole state below the $N = 161$ subshell. Consequently, characterizing the α-decay properties linking ^{273}Ds and ^{269}Hs provides a highly sensitive probe into the underlying $N = 162$ shell structure.

The isotope ^{273}Ds was initially and unambiguously identified at the GSI velocity filter SHIP as an α-decay product of cold fusion reactions [5]. Notably, in the ^{208}Pb(^{70}Zn, n)^{277}Cn cold fusion experiments conducted at GSI [5, 6] and RIKEN [7, 8], the measured lifetime of the ^{273}Ds descendant consistently fell within the 0.1 ms order of magnitude. Recently, the direct production of ^{273}Ds via the ^{238}U(^{40}Ar, $5n$) reaction was investigated by Oganessian et al. [9] Two decay chains were identified via established genetic correlations [10]; one sequence exhibited agreement with previous data, while the other was attributed to an isomeric state of ^{273}Ds. However, both observed decay chains were incomplete, providing limited information regarding the decay properties of the daughter nucleus ^{269}Hs. Subsequently, theoretical evaluations of these decay sequences were performed by Wang et al. [11] employing the density-dependent cluster model (DDCM+) and the unified decay formula (UDF), prompting minor proposed revisions to the assigned decay paths. In the present work conducted at the SHANS-2 facility [12], the same reaction was utilized, resulting in the observation of two additional, distinct, and complete decay paths. These comprehensive chains, rigorously verified through genetic correlations [10] in agreement with historical data, provide robust and independent confirmation for the existence of the two distinct states of ^{273}Ds, designated as ^{273}Ds^a and ^{273}Ds^b, as initially reported by Oganessian et al. [9].

In the present work, based on the distinct ^{273}Ds decay paths observed herein coupled with relevant historical data, we aim to investigate the existence of the ^{273}Ds isomeric state and explore the potential for isomerism

in ^{269}Hs. Furthermore, by incorporating macroscopic-microscopic models, we extract structural insights regarding the $N = 162$ neutron shell. The experiment is described below, and experimental data are presented.

2. Experiment and results

The experiment was performed at the China Accelerator Facility for Superheavy Elements (CAFE2) of the Institute of Modern Physics (IMP) using the gas-filled recoil separator SHANS2 (Spectrometer for Heavy Atoms and Nuclear Structure-2). The superheavy nucleus ^{273}Ds was synthesized via the ^{238}U(^{40}Ar, $5n$) fusion-evaporation reaction. A ^{40}Ar^{13+} beam, with incident average energies of 217, 220 and 225 MeV, was delivered by a superconducting linear accelerator. The typical beam intensity on target was approximately 5 pμA, accumulating a total effective irradiation time of roughly 728 hours.

The targets were prepared by magnetron sputtering ^{238}U onto 2.22 and 2.19 μg/cm^2 thick Ti backings, resulting in average ^{238}U layer thicknesses of 470 and 604 μg/cm^2. These arc-shaped target segments were mounted on a 50-cm-diameter wheel, which rotated at 1500 rpm during irradiation to withstand the intense beam bombardment.

Evaporation residues (ERs) were separated in-flight from the primary beam and other reaction byproducts using SHANS2. The separator was filled with helium gas at a pressure of approximately 1 mbar, with its magnetic rigidity optimized for ^{273}Ds transmission. At the focal plane, the ERs first passed through two multi-wire proportional chambers (MWPCs) before being implanted into a double-sided silicon strip detector (DSSD). The MWPCs, mounted about 20 cm upstream of the DSSD and operated with isobutane gas, and the MWPCs were primarily utilized to discriminate implantation events from decay events based on time–position correlations, combining time-of-flight (TOF) information with the triggering status of the MWPCs.

The DSSD has an active area of 128×48 mm^2 and a thickness of 300 μm. Its front side (the implantation face, defined as the Y-plane) is segmented into 48 horizontal strips, while the back side (the X-plane) is segmented into 128 vertical strips. This configuration forms a total of 6144 pixels, providing a spatial resolution of approximately 1 mm. To detect α particles and spontaneous fission (SF) fragments escaping from the front side of the DSSD, six single-sided silicon detectors (SSDs) were arranged in a box-like geometry upstream of the DSSD. Each SSD has a thickness of 500 μm and an active area of 120×63 mm^2. Furthermore, three 300-μm-thick silicon detectors were mounted immediately behind the DSSD to serve as Veto detectors, thereby rejecting signals from light particles punching through the focal-plane detectors.The entire silicon detector array was cooled to approximately -30°C using a circulating liquid alcohol system to suppress thermal noise and improve energy resolution. For full-energy α particles in the 5–9 MeV range, the typical energy resolution (FWHM) of the DSSD was about 30 keV. The detection efficiency for α particles was approximately 87(6)%. All detector signals, after preamplification and shaping, were acquired by 14-bit analog-to-digital converters (ADCs) operating at a 100 MHz sampling rate in a triggerless mode. Further details regarding the SHANS2 separator and the detection system can be found in Refs. [12, 13].

In this work, the silicon detectors were calibrated using standard α sources(^{239}Pu, ^{241}Am, and ^{244}Cm) and the well-known α-decay lines of ^{218}Rn [14], ^{214}Po [15],^{216}Po [16], and ^{217}At [17], with reference energies taken from the evaluated Nuclear Data Sheets.To deduce the cross-sections, simulations were performed following the method detailed in Ref. [18], yielding a transmission efficiency of 20%. The experimental conditions and measured cross sections are summarized in Table 1.

Table 1: The ^{238}U target thicknesses; average laboratory-frame projectile energies $\bar{E}_{\rm lab}$ at the middle of the target layers; corresponding average excitation energies $\bar{E}^*$ (with use of mass tables [19, 20]) of the compound nucleus; total beam doses; and the measured cross sections σ (or upper limits) for the $4n$- and $5n$-evaporation channels in the ^{40}Ar + ^{238}U reaction. Values separated by "|" denote measurements performed with two different target thicknesses; the corresponding beam doses are listed in the same order.

Target thickness (mg/cm^2)	$\bar{E}^a_{\rm lab}$ (MeV)	$\bar{E}^*$ (MeV)	Beam dose ($\times 10^{19}$)	σ_{4n} (pb)	σ_{5n} (pb)
0.47|0.60	225	50	2.28|2.96	< 0.15	$0.14^{+0.18}_{-0.08}$
0.47	220	45	0.817	< 1.1	< 1.1
0.60	217	42	2.04	< 0.34	< 0.34

[a] The beam energy was measured with a systematic uncertainty of 1 MeV.

3. Results and discussion

For the ^{238}U(^{40}Ar, xn)$^{278-x}$Ds reaction at an excitation energy of approximately 50 MeV, the $5n$ evaporation channel is the most probable[9]. At this excitation energy, two decay chains were observed, the complete details of which are illustrated in Fig. 1. The energy and decay time of α_1 in Chain 1 are consistent with those reported in previous cold-fusion ^{208}Pb(^{70}Zn, n)^{277}Cn experiments at GSI [5, 6] and RIKEN [7, 8], as well as the hot-fusion ^{238}U(^{40}Ar, $5n$)^{273}Ds experiment at JINR [9]. Furthermore, the subsequent decay properties of ^{269}Hs (producing α_2 and α_3) and the terminal spontaneous fission (SF) of ^{261}Rf in both chains align with decay data documented in Refs. [3, 21–26]. Detailed assignments are discussed as follows.

Düllmann et al. [21] explicitly proposed that ^{261}Rf, populated via the α decay of ^{265}Sg, exhibits two states: ^{261}Rf^a and ^{261}Rf^b. The existence and corresponding decay properties of these two states were subsequently confirmed

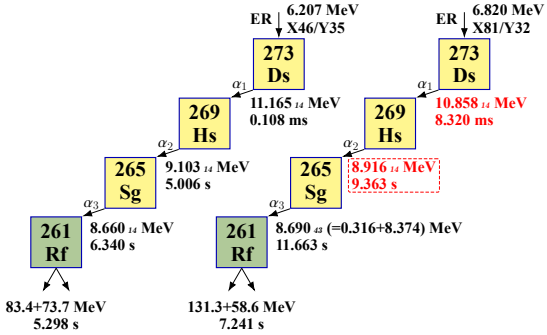

Figure 1: Two decay chains assigned to ^{273}Ds observed in the ^{238}U(^{40}Ar, $5n$)^{273}Ds reaction. For each chain, the energy of the evaporation residue (ER) event and its X/Y position on the detector are given at the top, followed by the energies of the successive α-decay events and the terminal spontaneous-fission (SF) event, together with the time intervals between adjacent events. Energies in parentheses denote summed signals. Italic values indicate the corresponding energy uncertainties. Red text and red-framed entries highlight the information validated in the present work and the inferred assignments related to the proposed isomeric states.

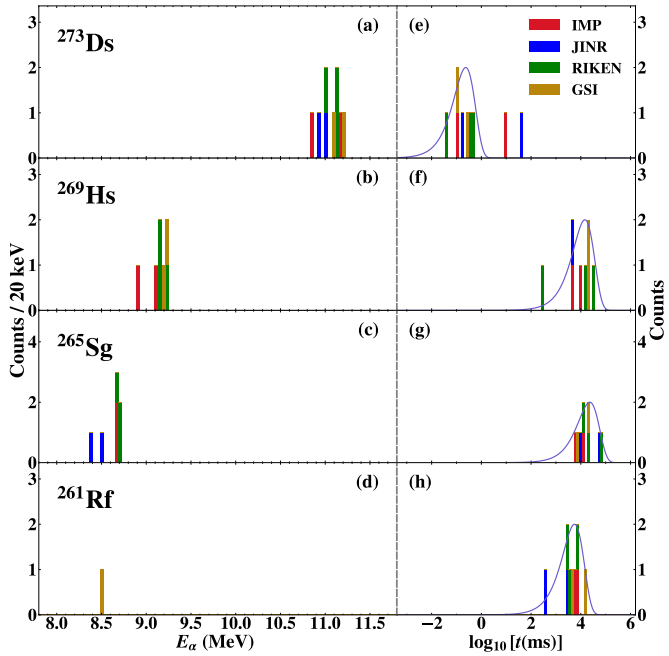

Figure 2: Comparison of the E_α and half-life information for decay chains involving ^{273}Ds. The red, blue, green, and yellow symbols correspond to the IMP, JINR[9], RIKEN[7, 8], and GSI[5, 6]. The fit curves in the decay-time panels are drawn according to Ref. [27].

by Haba et al. [25] via the ^{248}Cm(^{22}Ne, $5n$)^{265}Sg reaction. Specifically, ^{261}Rf^a decays to ^{257}No by emitting an 8.28MeV α particle with a half-life ($T_{1/2}$) of 68 s, whereas ^{261}Rf^b decays with $T_{1/2}$ = 2.6 s, exhibiting an 18% α-decay branch (E_α = 8.51 MeV) to ^{257}No and an 82% SF branch[25]. Evidently, the terminal SF events in both chains observed in the present work originate from the ^{261}Rf^b state. Furthermore, Düllmann et al. [3] first proposed that the α decay of ^{269}Hs proceeds via two distinct α-particle energies (E_α = 9.13 and 8.95 MeV), populating two corresponding states in the daughter nucleus, ^{265}Sg^a and ^{265}Sg^b. These two energies perfectly match the E_{α_2} values observed in Chain 1 and Chain 2, respectively. The decay properties of ^{265}Sg^a and ^{265}Sg^b were later established by Haba et al. [26] using high-statistics data from the ^{248}Cm(^{22}Ne, $5n$)^{265}Sg reaction. They determined that ^{265}Sg^a ($T_{1/2}$ = 8.5 s) decays via an 8.84 MeV α emission, feeding ^{261}Rf^a with a 91% branching ratio and ^{261}Rf^b with a 9% branching ratio. Conversely, ^{265}Sg^b ($T_{1/2}$ = 14.4 s) decays via an 8.69 MeV α emission, feeding ^{261}Rf^a and ^{261}Rf^b with branching ratios of 20% and 80%, respectively. In the present work, both observed chains are assigned to the ^{265}Sg^b $\rightarrow$ ^{261}Rf^b decay sequence. In summary, the E_α and $T_{1/2}$ values for all decay chains involving ^{273}Ds are compiled and compared with the historical data in Fig. 2.

3.1. Decay properties of ^{273}Ds

As illustrated in Fig.2, the majority of the data for Chain 1 and Chain 2 in the present work are consistent with historical data. However, the previously reported values for ^{273}Ds from Ref.[9] [E_α = 10.929(14) MeV, τ = 41.703 ms] and the values observed in this work (E_α = 10.858(14) MeV, τ = 8.320 ms) differ significantly from the α-decay energies and lifetimes measured at GSI and RIKEN. This discrepancy provides compelling evidence for the existence of a possible long-lived, low-decay-energy isomeric state in ^{273}Ds.

As previously noted, ^{273}Ds can be populated via two mechanisms: as an α-decay daughter of ^{277}Cn synthesized in cold-fusion ^{208}Pb(^{70}Zn, n) reactions at GSI [5, 6] and RIKEN [7, 8], or directly via hot-fusion ^{238}U(^{40}Ar, $5n$) reactions at JINR [9]. Currently, two distinct α-decay energies and lifetimes are observed for ^{273}Ds. Following the nomenclature in Ref. [9], the state characterized by the emission of lower-energy α particles and a longer lifetime is denoted as ^{273}Ds^a, while the state associated with higher-energy α emission is denoted as ^{273}Ds^b. It is noteworthy that ^{273}Ds populated via the α decay of ^{277}Cn exclusively occupies the ^{273}Ds^a state, whereas direct production populates both states with approximately equal probability. This observation further corroborates the assignment of ^{273}Ds^a as a possible isomeric state.

3.2. Decay properties of ^{269}Hs

Türler et al. [28] previously pointed out that all α-decay energies of ^{269}Hs (9.17–9.25 MeV) observed as the decay daughter of ^{277}Cn and ^{273}Ds are among the highest reported for this nuclide. They proposed two potential explanations: either energy summing with conversion electrons occurs due to implantation into a focal-plane detector, or long-lived isomeric states exist in ^{269}Hs, similar to those in ^{265}Sg and ^{261}Rf. Combining this context with the ^{269}Hs decay information obtained in the present work, a revised decay scheme is presented in Fig. 3. Adopting the nomenclature of Oganessian et al. [9], the state

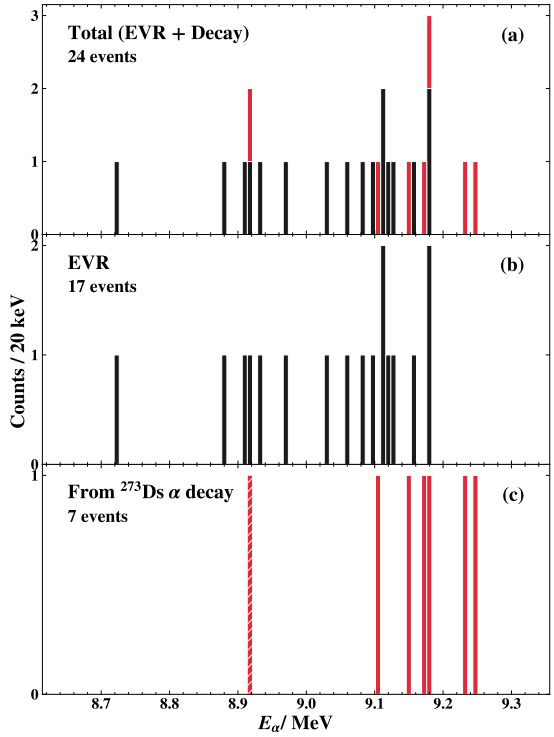

Figure 3: Energy distributions of α particles assigned to ^{269}Hs. The upper panel shows all events, the middle panel shows nuclei produced directly as evaporation residues (EVRs), and the lower panel shows nuclei populated through the α decay of ^{273}Ds. The hatched bar indicates the event observed in the present work at E_α = 8.915 MeV.

with E_α = 9.20(4) MeV and $T_{1/2} = 13^{+10}_{-4}$ s is denoted as ^{269}Hs^a, and the state with E_α = 9.08(15) MeV and $T_{1/2} = 2.8^{+13.6}_{-1.3}$ s is denoted as ^{269}Hs^b. In the present work, the state characterized by E_α = 8.915 MeV and τ = 9.363 s is assigned as ^{269}Hs^c. Due to the limited energy resolution in some historical data, distinguishing between ^{269}Hs^a and ^{269}Hs^b can be challenging; therefore, they are collectively referred to as ^{269}Hs^{a+b} where appropriate. As shown in Fig. 3, ^{269}Hs produced directly as evaporation residues (EVRs) is distributed across all three states. Using an energy boundary of 9.0 MeV, the population ratio is $n(^{269}\text{Hs}^{a+b}) : n(^{269}\text{Hs}^c) = 11 : 5$. In contrast, for ^{269}Hs populated via the α decay of ^{273}Ds, this ratio is $n(^{269}\text{Hs}^{a+b}) : n(^{269}\text{Hs}^c) = 6 : 1$. Specifically, the source of ^{269}Hs^c is the aforementioned possible isomeric state ^{273}Ds^a, yielding a ratio of $n(^{273}\text{Ds}^b \rightarrow {}^{269}\text{Hs}^a) : n(^{273}\text{Ds}^a \rightarrow {}^{269}\text{Hs}^c) = 1 : 6$. Notably, cross-decay scenarios such as ^{273}Ds$^b \rightarrow {}^{269}$Hs^c and ^{273}Ds$^a \rightarrow {}^{269}$Hs^a were not observed. Such strong decay selectivity strongly suggests the existence of ^{269}Hs^c as a possible isomeric state in ^{269}Hs, pending further theoretical support.

4. Discussion

For clarity, the tentative state assignments adopted in the present work are summarized in Table 2. On the basis of the present data, two distinct states are proposed for ^{273}Ds, denoted as ^{273}Ds^a and ^{273}Ds^b. By combining the present results with the data reported by Oganessian et al. [9], three groups are tentatively proposed for ^{269}Hs, denoted as ^{269}Hs^a, ^{269}Hs^b, and ^{269}Hs^c. At the present stage, the distinction between ^{269}Hs^a and ^{269}Hs^b remains less secure than that of ^{269}Hs^c, because of the limited statistics and the finite energy resolution of part of the historical data. Therefore, the present classification should be regarded as a working scheme for discussing the decay pattern and the underlying structure, rather than as a final level assignment. A more definitive conclusion will require higher-statistics measurements in future direct-production experiments.

Table 2: Tentative state assignments adopted in the present work. The numerical values can be updated after the final evaluation of all decay chains.

Nuclide	Decay mode	E_α (MeV)	Q_α (MeV)	$T_{1/2}$	Reference
^{273}Ds^a	α	[fill]	[fill]	[fill]	[fill]
^{273}Ds^b	α	[fill]	[fill]	[fill]	[fill]
^{269}Hs^a	α	[fill]	[fill]	[fill]	[fill]
^{269}Hs^b	α	[fill]	[fill]	[fill]	[fill]
^{269}Hs^c	α	[fill]	[fill]	[fill]	[fill]

The most direct experimental result of the present work is the emergence of two distinct decay patterns in ^{273}Ds. One branch is characterized by a higher α-decay energy and a shorter lifetime, whereas the other shows a lower α-decay energy and a significantly longer lifetime. When the present chains are considered together with the chains reported in Ref. [9], this separation is difficult to reconcile with a single-state interpretation. At the same time, the decay pattern of ^{269}Hs does not appear to be random. The high-energy and low-energy groups are not populated with the same weight in direct production and in the decay of ^{273}Ds, and no obvious cross feeding between the proposed ^{273}Ds^a and ^{269}Hs^{a,b} branches, or between ^{273}Ds^b and ^{269}Hs^c, is observed in the present data set. Such a selective decay pattern is naturally suggestive of configuration-dependent α decay. The first branch,which is strated with a higher α-decay energy and a shorter lifetime about ^{273}Ds,is assigned to

$$^{273}\text{Ds}^b \xrightarrow{\alpha} {}^{269}\text{Hs}^{a,b} \xrightarrow{\alpha} {}^{265}\text{Sg}^b \xrightarrow{\alpha} {}^{261}\text{Rf}^b \xrightarrow{\text{SF}} \text{fission}$$

And the second branch, which is characterized by a lower α-decay energy and a longer lifetime about ^{273}Ds, is assigned to

$$^{273}\text{Ds}^a \xrightarrow{\alpha} {}^{269}\text{Hs}^c \xrightarrow{\alpha} {}^{265}\text{Sg}^b \xrightarrow{\alpha} {}^{261}\text{Rf}^b \xrightarrow{\text{SF}} \text{fission}$$

A plausible physical interpretation is that the long-lived branch of ^{273}Ds is associated with a structure different from that of the short-lived branch. In a strongly deformed odd-N nucleus, the α-decay rate is controlled not only by Q_α, but also by the change of angular momentum, the overlap between the initial and final intrinsic configurations,

and the corresponding hindrance of α formation and barrier penetration[29]. Therefore, a marked difference in lifetime between two branches of the same nuclide can arise even when the difference in Q_α is modest. In this context, the observed separation between ^{273}Ds^a and ^{273}Ds^b is consistent with the presence of two competing quasineutron configurations, one of which gives rise to a hindered, longer-lived α decay.

This interpretation becomes particularly interesting in view of the expected shell structure around the deformed $N = 162$ gap [30]. The nucleus ^{273}Ds contains one neutron particle above this gap, whereas ^{269}Hs corresponds to one neutron hole below it. The decay chain ^{273}Ds $\rightarrow$ ^{269}Hs therefore probes the single-particle structure on both sides of the same deformed shell closure. Theoretical calculations indicate that several low-lying high-Ω neutron configurations may be relevant in this region. In particular, the unique-parity orbital $\nu 13/2^-[716]$ has been predicted to lie close to the Fermi surface, while macroscopic–microscopic calculations for neighboring odd-N transfermium nuclei also place $\nu 11/2^-[725]$ among the low-lying candidates [31, 32]. Within such a picture, it is reasonable to consider whether the long-lived branch of ^{273}Ds may be associated with a high-Ω one-neutron configuration. This possibility is qualitatively consistent with the occurrence of high-K isomerism in the neighboring deformed superheavy region, as discussed for the ^{270}Ds $\rightarrow$ ^{266}Hs decay sequence [33]. At present, however, the available data do not allow a unique spin-parity or Nilsson-orbital assignment; both $\nu 13/2^-[716]$ and $\nu 11/2^-[725]$ related interpretations or other possibilities remain open.

Under this interpretation, the decay selectivity observed in the daughter nucleus ^{269}Hs also acquires a natural explanation. If ^{273}Ds^a and ^{273}Ds^b correspond to different intrinsic configurations, then favored α decay is expected to populate daughter states with similar structure. The proposed ^{269}Hs^c state, which appears in the low-energy branch and is preferentially associated with ^{273}Ds^a, may therefore reflect a corresponding low-lying configuration in the daughter nucleus. By contrast, the higher-energy groups ^{269}Hs^a and ^{269}Hs^b may be associated with the decay of ^{273}Ds^b. In this sense, the present data may indicate not only a possible isomeric structure in ^{273}Ds, but also a correlated multistate structure in ^{269}Hs. It should be emphasized, however, that the present evidence for three distinct states in ^{269}Hs is still tentative, and the possibility that part of the observed structure reflects unresolved subgroups cannot yet be excluded.

Additional support for this picture may be obtained from the isotopic systematics. As shown in Fig. 4, the evolution of Q_α, half-life, and reduced α-decay width along the Ds and Hs isotopic chains provides a broader context for the present assignments. In a simple single-configuration picture, these observables are expected to evolve rather smoothly with neutron number. A local separation of branches, especially when it appears simultaneously in decay energy, lifetime, and reduced width, is more naturally interpreted as a signature of a structural change than as a mere statistical fluctuation. In the present case, the long-lived branch of ^{273}Ds and the low-energy branch of ^{269}Hs stand out from the local trends and therefore strengthen the interpretation in terms of different underlying configurations. At the same time, it should be noted that the reduced width in this strongly deformed superheavy region is model dependent, because deformation, angular-momentum change, and structure effects all enter the extraction. For this reason, the reduced width should not be regarded here as a standalone proof of isomerism. Its main value in the present work is comparative: when evaluated consistently within the same prescription, it provides an additional indicator that the proposed branches are structurally different.

The excitation-function information gives a further consistency check for the present interpretation. As shown in Fig. 5, the experimental cross sections are broadly consistent with the Hivap expectation that the $5n$ channel dominates near the present beam-energy setting. This agreement supports the assignment of the observed chains to ^{273}Ds produced in the ^{238}U(^{40}Ar,$5n$)^{273}Ds reaction, rather than to neighboring evaporation channels. Although the excitation function by itself does not distinguish between different intrinsic states, it does show that the observed long-lived and short-lived branches emerge within the same reaction framework. This point is important, because it indicates that the difference between the two branches is more likely to be structural than to originate from a different reaction channel.

In summary, the present data support a picture in which ^{273}Ds exhibits two distinct decay branches and ^{269}Hs may contain at least one additional low-energy state beyond the previously recognized high-energy groups. A consistent interpretation of these observations is that the odd-N nuclei around the deformed $N = 162$ shell closure display configuration-dependent α decay, possibly involving a high-Ω structure related to the $\nu 13/2^-[716]$ orbital. If confirmed by future high-statistics experiments and dedicated structure calculations, this would extend the evidence for low-lying isomeric structures in the transfermium region and provide new constraints on the single-particle level ordering near the deformed shell gap at $N = 162$.

5. Summary and conclusions

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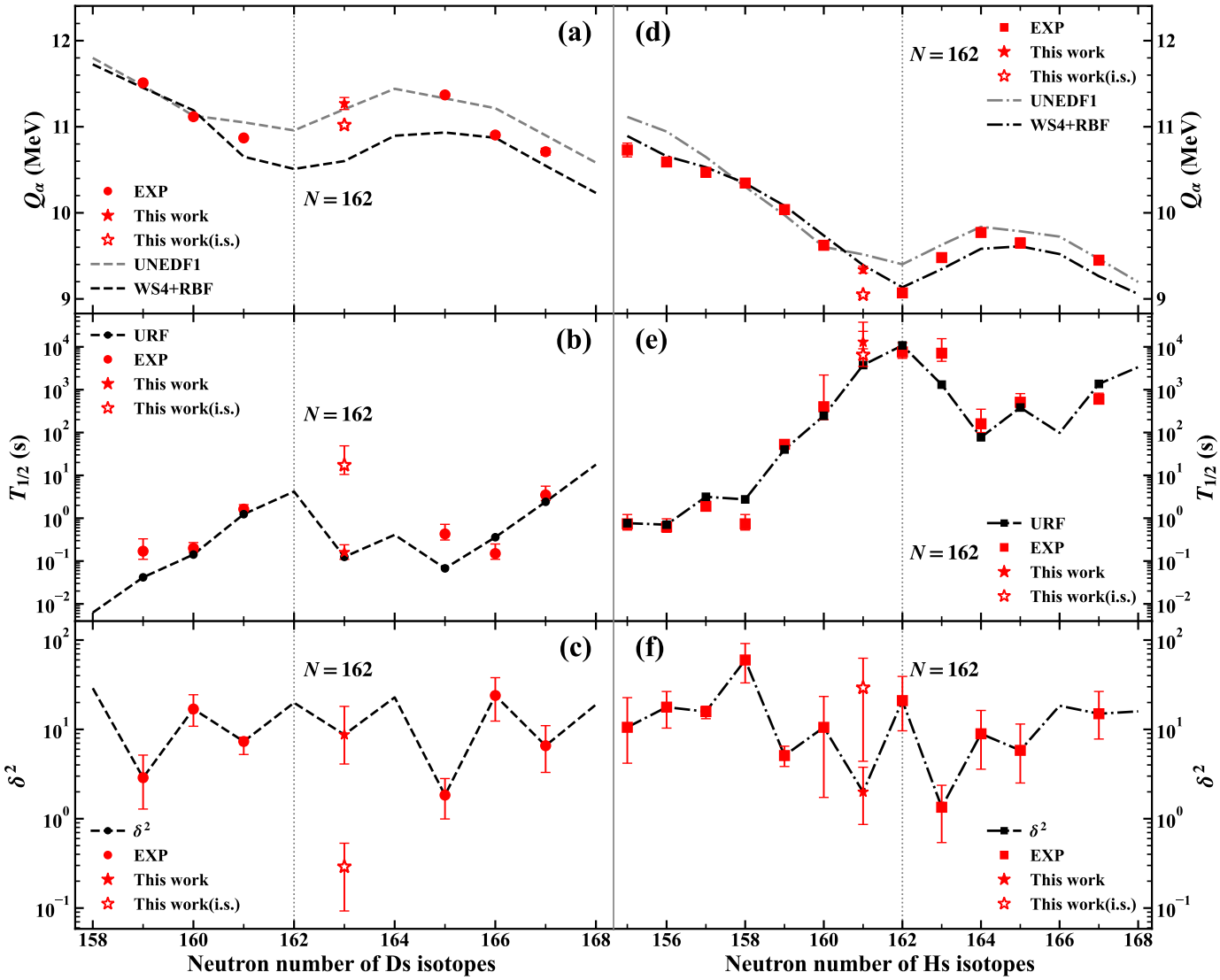

Figure 4

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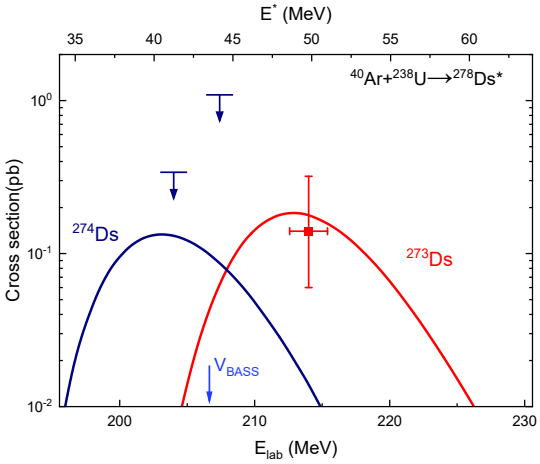

Figure 5: Energy distributions of α particles assigned to ^{269}Hs. The upper panel shows all events, the middle panel shows nuclei produced directly as evaporation residues (EVRs), and the lower panel shows nuclei populated through the α decay of ^{273}Ds. The hatched bar indicates the event observed in the present work at E_α = 8.915 MeV.

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Acknowledgements

Thanks to ...

Appendix A. Appendix title 1

Appendix B. Appendix title 2

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